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Project Title:	The CQ Experiment		
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Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
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Zip Code:	91125	**Congressional District:**	29
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Duncan, Robert (University of New Mexico) Chui, Talso (JPL) Day, Peter (JPL) Lee, Richard (Caltech)		
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Task Description:

The heat capacity of superfluid helium in a constant heat flux has been predicted to become infinite at a critical value of the heat flux where superflow becomes unstable. Measurements done in our laboratory at Caltech have succeeded in detecting an excess heat capacity due to the presence of a heat flux but, for a variety of gravity-related reasons, it is impossible in the laboratory to make measurements at values of the heat flux that approach the critical value. The CQ Experiment will measure the heat capacity (C) at constant heat flux (Q) in the absence of gravity.
Objective:

The CQ Experiment will be a guest experiment, using the hardware of Critical Dynamics in Microgravity (DYNAMX) without engineering modification. The DYNAMX Instrument Sensor Package (ISP) was not designed for precision calorimetry, but is nearly ideal for that purpose. Scheduled for the M1 mission aboard the Low Temperature Microgravity Physics Facility (LTMPF), DYNAMX should require about one month to complete its measurements. The liquid helium on the M1 mission is expected to last 4-5 months. The CQ experiment will be done in the DYNAMX ISP during the remaining time.

The CQ Experiment will measure the heat capacity of superfluid helium at constant heat flux, over a range of temperatures within one millionth of a Kelvin of the superfluid transition (the lambda point, where helium becomes superfluid in the absence of a heat flux). The range of heat fluxes will be from zero up to a few millionths of a Watt per square centimeter. Only in these conditions can helium conduct heat at constant temperature (within the 50 pico-Kelvin resolution of our thermometry). At higher heat fluxes, temperature gradients would render heat capacity measurements meaningless. With these small heat fluxes the experimental signal, excess heat capacity due to the heat flux, becomes resolvable only within one micro-Kelvin of the lambda point.

The goal of the experiment is to measure the heat capacity as close as possible to the critical value of the heat flux, where superflow becomes unstable, and where the heat capacity is predicted to approach infinity.

On Earth, the hydrostatic pressure of a column of liquid helium causes the lambda point temperature itself to vary by more than one micro-Kelvin per centimeter of column height. To combat this problem, our laboratory measurements were made in a cell only 0.06 centimeters high. Even so, all our data are severely compromised by the hydrostatic gravity effect. In addition, the very small cell height made it necessary to rely on temperatures measured in the end caps of the cell, greatly complicating data analysis and limiting the range of measurements to values of the heat flux far below the critical value. All or most of these problems will be overcome by making the measurements in the DYNAMX ISP. The DYNAMX cell is 11 mm high and measures the temperature with high resolution thermometry at three positions along the side wall.

Significance:

There are two outstanding discrepancies in this field at the moment. The first is that experiment and theory disagree on the value of the critical heat current as a function of temperature, and the other is that the excess heat capacity measured in our own experiments is much larger than that predicted theoretically. A successful CQ experiment would shed considerable light on the causes of both of these discrepancies. If in addition it turns out to be possible to measure the heat capacity close enough to the critical heat current to observe the functional form of the predicted divergence, that result might very well open a whole new field of physics. The form of the divergence should be dominated by fluctuations of a qualitatively different kind from those that govern the behavior of conventional critical points.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Enhancing America's scientific/technological standing by exploring the frontiers of physics obscured by Earth's gravity. Innovative technologies to be reported in a later funding year.

Since the cancellation of the DYNAMX/CQ flight projects, in Sept. 2004 due to NASA's new Moon and Mars initiative, our two teams have had to consolidate resources. To this end some of the DYNAMX/CQ ground science has moved from the University of New Mexico (UNM) to Caltech. We have two experiments at Caltech focusing on different aspects of the CQ science. Our first experiment expands on our previous measurements of the heat capacity of liquid helium as it transitions into the Self Organized Critical (SOC) state [R. A. M. Lee et al., JLTP, 134, Nos. ½, 495-505 (2004)] - this is a unique state in which the helium sample exists in steady-state at its critical point, T_SOC(Q), under an applied heat flux, Q. It is hoped that these measurements will elucidate the physics of the CQ experiment as it approaches its critical point, T_C(Q).
This experiment has been successful in measuring, for the first time, the heat capacity of the fully developed SOC state. Since the SOC state matches the gravity-induced temperature gradient in the superfluid transition temperature, T_lam, it is unaffected by any hydrostatic gravity rounding. Therefore, uniquely for a ground experiment, the SOC heat capacity closely matches the zero-gravity, static, heat capacity as measured by previous flight experiments [The Lambda Point Experiment]. This work was presented at the 24th International Conference on Low Temperature Physics (2005) [Chatto, A. R. et al., AIP Conf. Proc. 850, 137 (2006)].

We have also expanded and improved upon previous measurements [D. A. Sergatskov et al., JLTP, 134, Nos. ½, 517-526 (2004)] of a new temperature-entropy wave that propagates against the applied heat flux when the helium sample is in the SOC state. This data is in remarkable agreement with an improved theoretical treatment of the SOC state [Chatto A. R., Ph.D. thesis, Caltech, 2006] that treats the helium as a normal fluid with a depressed superfluid transition temperature at T_SOC(Q). This work was presented at the International Symposium on Quantum Fluids and Solids (2006).

The second experiment is contained within a pre-existing cryo-probe that was transported from UNM to Caltech in September, 2004. This apparatus contains all of the thermal isolation and control technology that has been designed for the flight experiment, including, at its heart, the experimental cell that has been constructed to flight specifications. In order to ameliorate the effect of gravity this cell has been made shorter than the flight cell and contains only two sidewall thermometer probes. In addition we have added a bottom endplate thermometer probe to enable new measurements of the singular Kaptiza resistance - see below.

This new cell has now been fully integrated into the cryo-probe and tested. Our main emphasis, from a flight perspective, was to test the impact on the science measurements due to one part of the cell, the "bubble chamber" - this was an integral part of the flight apparatus, required as a safety measure in the original flight mission. Unfortunately,

Task Progress:	our evaluation testing did reveal an adverse, unforeseen, effect on the CQ science measurements and it was decided to remove the bubble chamber and revert back to a proven ground-based design; our main emphasis being to obtain science data. This important testing will, however, aid DX/CQ in a re-design of the bubble chamber should this project be re-selected for flight. Our present status is that all modifications to the cell have been completed. The experiment has been re-integrated into the cryo-probe and is now functioning, after an elusive leak on the sidewall was found and repaired. Unfortunately, the high resolution magnetic susceptibility thermometers (capable of 50pK temperature resolution) exhibit excessive temperature drift. The source of this drift must first be found and fixed before precise heat capacity data can be taken. With this apparatus the CQ/DX teams hope to make measurements of the following: 1. new measurements of CQ - the heat capacity under a constant heat flux, Q. This follows recent thermal modelling, conducted primarily by Prof. Stephen Boyd (UNM), which suggests that inhomogeneous heat flow could amplify the theoretically predicted enhancement and may have lead to the results reported by Harter A. et al., Phys. Rev. Lett., 84, 2195 (2000). This work was presented at the 24th International Conference on Low Temperature Physics (2005) [Boyd S. T. P. et al., AIP Conf. Proc. 850, 133 (2006)]. The new CQ/DX cell ameliorates inhomogeneous heat flow by a novel, new, design of endplate that serves to spread the heat flux evenly before it enters the helium sample. 2. new measurements of the normal-fluid/superfluid interface advance rate, as a function of various experimental parameters in this 'Heat from Below' geometry. From this we hope to infer information on the thermal conductivity of liquid helium, exceptionally close to the superfluid transition. 3. new measurements of the superfluid thermal resisitivity, induced by mutual friction effects, will be made and compared to the 'Heat from Above' results reported by Lee R. A. M. et al., JLTP, 134, Nos. ½, 495-505 (2004). This will allow us to experimentally determine the geometry-dependent effect on mutual friction near the superfluid transition. 4. new measurements of the singular Kapitza resistance and the temperature, T_DAS(Q), at which superfluidity breaks down at the heated bottom endplate. From these new, high fidelity, measurements we hope to reconfirm our previous ground-based measurements, including those of other authors, and clarify some of the discrepancies between those previous measurements and theory. Once this is accomplished these measurements may very well represent just about the best that can be achieved on Earth under the deleterious effect of gravity.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Chatto AR, Lee RAM, Duncan RV, Day PK, Goodstein DL. "Measurement of the SOC state specific heat in 4He." AIP Conference Proceedings. 2006 Sep;850:137-8. (24th International Conference on Low Temperature Physics). http://dx.doi.org/10.1063/1.2354640 , Sep-2006
Articles in Peer-reviewed Journals	Boyd STP, Chatto AR, Lee RAM, Duncan RV, Goodstein DL. "Effect of inhomogeneous heat flow on the enhancement of heat capacity in Helium-II by counterflow near T_lamda." AIP Conference Proceedings. 2006 Sep;850:133-4. (24th International Conference on Low Temperature Physics). http://dx.doi.org/10.1063/1.2354638 , Sep-2006
Dissertations and Theses	Chatto AR. "Experiments on the Self Organized Critical State of He-4." Dissertation, Ph.D., Caltech, Pasadena, August 2006. , Aug-2006